Lab Mid Evaluation Report

Course Code : CSE 412

Course Title : Computer Graphics Lab

Experiment No : 01

Experiment Name : Drawing a Trapezium Using DDA and Bresenham Algorithm

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Lab Mid Evaluation

Drawing a Trapezium Using DDA and Bresenham Algorithm

Title: Drawing Multiple Polygons Using OpenGL

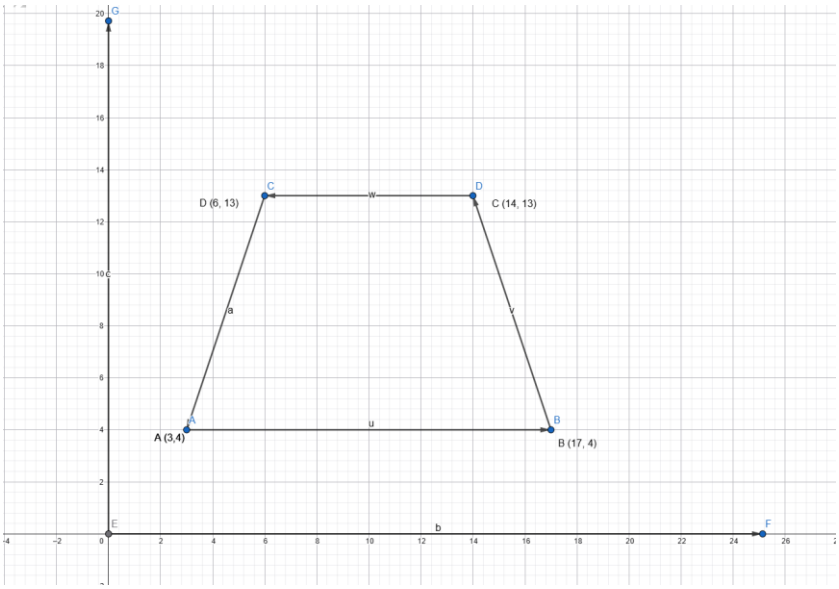
Introduction:

In this project, a trapezium shape is drawn using line drawing algorithms. Bresenham and DDA algorithms are used to draw the sides of the shape.

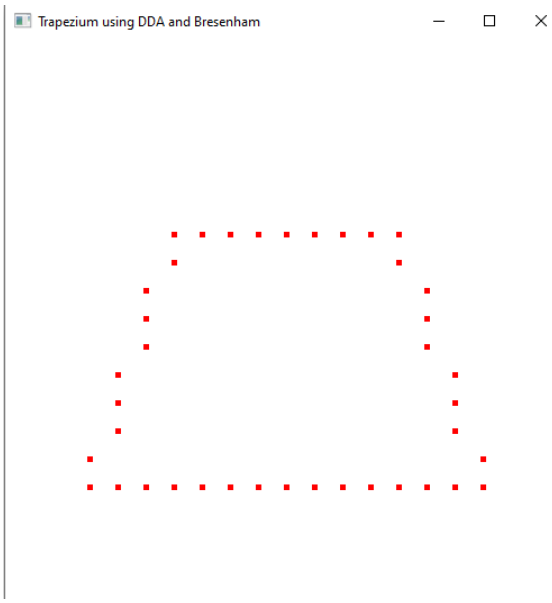
Contents:

- **Functions Used:** `drawLineDDA(),drawLineBresenham(),glVertex2i(),glBegin()`
- **Shape Used:** Trapezium (using four lines)

Graph:



Draw X and Y axis. Mark four points and connect them to form a trapezium.



Code (Drawing Portion Only):

```
void drawLineDDA(int x1, int y1, int x2, int y2)
{
    float dx = x2 - x1;
    float dy = y2 - y1;

    float steps;

    if (fabs(dx) > fabs(dy))
        steps = fabs(dx);
    else
        steps = fabs(dy);

    float Xinc = dx / steps;
    float Yinc = dy / steps;
```

```
    float x = x1;
    float y = y1;

    glBegin(GL_POINTS);

    for (int i = 0; i <= steps; i++)
    {
        glVertex2i(round(x), round(y));
        x = x + Xinc;
        y = y + Yinc;
    }

    glEnd();
}



void drawLineBresenham(int x1, int y1, int x2, int y2)
{
    int dx = abs(x2 - x1);
    int dy = abs(y2 - y1);

    int sx = (x1 < x2) ? 1 : -1;
    int sy = (y1 < y2) ? 1 : -1;

    int err = dx - dy;

    glBegin(GL_POINTS);
```

```
    while (true)
    {
        glVertex2i(x1, y1);

        if (x1 == x2 && y1 == y2)
            break;

        int e2 = 2 * err;

        if (e2 > -dy)
        {
            err = err - dy;
            x1 = x1 + sx;
        }

        if (e2 < dx)
        {
            err = err + dx;
            y1 = y1 + sy;
        }
    }

    glEnd();
}
```

Discussion:

The task was to draw a trapezium. Three sides were drawn using Bresenham algorithm and one side was drawn using DDA algorithm.